Beyond One Course: Do Behavioral-State Classifiers Transfer Across Programming Courses?

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Abstract

Classifiers of learner behavioral states are almost always trained and evaluated within a single course. We trained a gradient-boosted classifier on labeled windows from a Python CS1 course and evaluated it on two other courses: a Java data structures course and a block-based CS0 course. Macro-F1 fell from 0.72 in-course to 0.61 and 0.55. Per-course normalization of rate features recovered part of the gap (0.66 on data structures). Errors concentrated on the "stuck" state, whose behavioral signature depended on the language's error messages. We recommend reporting cross-course results as a standard check.

1 Introduction

A behavioral-state classifier that works in one course is a finding about that course. Whether it is a finding about learners depends on transfer. We test transfer across three courses that differ in language, level and editor.

2 Data

Course	Students	Labeled windows	Language
CS1 (train)	140	3,050	Python
Data structures	62	1,120	Java
CS0	88	980	Blocks

Windows were 30 seconds long and labeled with a four-state scheme (productive, stuck, help-seeking, off-task) by raters trained on the same guide. Within-course agreement was kappa 0.68, 0.65 and 0.59.

3 Results

Test course	Macro-F1 (raw)	Macro-F1 (normalized)	F1 stuck
CS1 (held-out students)	0.72	0.73	0.69
Data structures	0.61	0.66	0.48
CS0	0.55	0.57	0.41

Normalizing edit and run rates by each course's median helped most where students' baseline pace differed. The stuck state transferred worst. In Java, compiler errors arrive before a run and stuck students compiled repeatedly; in CS0, blocks prevent most syntax errors and stuck students instead dragged blocks in and out without running.

4 Recommendations

• Report at least one cross-course or cross-term evaluation.

• Normalize rate features per course before training.

• Expect "stuck" to need course-specific features tied to the error channel.

5 Limitations

Three courses at two institutions; label schemes were harmonized, which may hide real differences in what raters meant.

References

[1] Haldane, I., Okafor, T., and Brandt, L. Replaying the keystroke. SIGCSE TS 2024, 512-518.

[2] Ferreira, A. and Kowalczyk, P. Group-aware splits for educational sequence data. JEDM 15, 2 (2023).